If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.

b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.

c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.

d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).

e. Bonus problems are not required for credit, but they are worth extra credit.

Problems

Problem 1. (Polynomial Kernel.) Let the feature map be

$$\phi(x) = (1, x, x^2)^\top.$$

(a.) Write the corresponding kernel $K(x, y) = \phi(x)^\top \phi(y)$.

(b.) Compute $K(1, 2)$ and $K(0, 3)$.

(c.) Is this kernel linear, polynomial, or nonlinear? Explain.

Solution.
We have

$$k(x, x') = \phi(x)^\top \phi(x') = 1 + xx' + x^2 x'^2 = (1 + xx')^2 - xx'.$$

Hence:

$$k(1, 2) = 1 + 1 \cdot 2 + 1^2 2^2 = 1 + 2 + 4 = 7, \qquad k(0, 3) = 1 + 0 + 0 = 1.$$

This is a *polynomial kernel* of degree 2, since it contains terms up to $x^2 x'^2$. **Problem 2.** (Kernel Matrix Structure.) Let $x_1 = 0$, $x_2 = 1$, $x_3 = 2$ and $k(x, x') = (1 + xx')^2$.

(a) Compute the kernel matrix $K \in \mathbb{R}^{3\times 3}$.

(b) Compute $\operatorname{rank}(K)$.

Solution. $K_{ij} = (1 + x_i x_j)^2$. Thus

$$K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 9 \\ 1 & 9 & 25 \end{bmatrix}.$$

$\det(K) = 32 \neq 0$, so $\operatorname{rank}(K) = 3$. Hence the feature map lives in $\mathbb{R}^3$—it corresponds to $(1, \sqrt{2}\,x, x^2)^\top$.

Problem 3. (Primal–Dual Consistency.) Show that the primal and dual kernel ridge regression predictions are identical: that is, show that

$$\Phi\hat{\theta} = K\hat{\alpha}, \quad \text{where } \hat{\theta} = (\Phi^\top\Phi + \lambda I)^{-1}\Phi^\top y,\ \hat{\alpha} = (K + \lambda I)^{-1}y,\ K = \Phi\Phi^\top.$$

Solution. Multiply the first equation by Φ:

$$\Phi\hat{\theta} = \Phi(\Phi^\top\Phi + \lambda I)^{-1}\Phi^\top y.$$

Using the matrix identity $\Phi(\Phi^\top\Phi + \lambda I)^{-1}\Phi^\top = (\Phi\Phi^\top)(\Phi\Phi^\top + \lambda I)^{-1} = K(K + \lambda I)^{-1}$, we get $\Phi\hat{\theta} = K\hat{\alpha}$.

Problem 4. (Centered Kernel Matrix.)
Let K be any $n \times n$ kernel matrix and $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. The centered kernel matrix is $K_c = HKH$.

(a) Show that $K_c\mathbf{1} = 0$ and $\mathbf{1}^\top K_c = 0^\top$.

(b) Show that for any vector $y \in \mathbb{R}^n$,

$$y^\top K_c y = (Hy)^\top K(Hy).$$

(c) Hence conclude that centering the kernel matrix is equivalent to replacing the features $\phi(x_i)$ by their mean-centered versions $\phi(x_i) - \bar{\phi}$, where $\bar{\phi} = \frac{1}{n}\sum_i \phi(x_i)$.

Solution.
Let $K \in \mathbb{R}^{n\times n}$ be any kernel matrix and let

$$H = I - \frac{1}{n}11^\top, \qquad 1 = (1, \dots, 1)^\top \in \mathbb{R}^n.$$

The centered kernel matrix is

$$K_c = HKH.$$

(a) **Show that** $K_c 1 = 0$ **and** $1^\top K_c = 0^\top$**.**

Compute

$$H1 = \Big(I - \frac{1}{n}11^\top\Big)1 = 1 - \frac{1}{n}1(1^\top 1) = 1 - \frac{1}{n}(n)1 = 1 - 1 = 0.$$

Similarly,

$$1^\top H = 1^\top - \frac{1}{n}(1^\top 1)1^\top = 1^\top - 1^\top = 0^\top.$$

Therefore,

$$K_c 1 = HKH\,1 = HK(0) = 0, \qquad 1^\top K_c = 1^\top HKH = 0^\top KH = 0^\top.$$

Thus every row and every column of K_c sums to zero.

(b) **Show that for all** $y \in \mathbb{R}^n$**,** $y^\top K_c y = (Hy)^\top K(Hy)$.

We compute

$$y^\top K_c y = y^\top HKHy = (Hy)^\top K(Hy),$$

using the fact that H is symmetric: $H^\top = H$.

(c) **Conclude that centering features is equivalent to centering the kernel.**

Assume the kernel arises from a feature map $\phi(x_i) \in \mathcal{H}$ in a Hilbert space:

$$K_{ij} = \langle \phi(x_i),\, \phi(x_j) \rangle.$$

Define the mean feature vector

$$\bar{\phi} = \frac{1}{n} \sum_{i=1}^{n} \phi(x_i), \qquad \tilde{\phi}(x_i) := \phi(x_i) - \bar{\phi}.$$

We now show that K_c is exactly the Gram matrix of the centered features $\tilde{\phi}(x_i)$.

For any $y \in \mathbb{R}^n$,

$$y^\top K y = \sum_{i,j} y_i y_j \langle \phi(x_i), \phi(x_j) \rangle = \Big\| \sum_{i=1}^{n} y_i \phi(x_i) \Big\|^2.$$

Part **(b)** shows

$$y^\top K_c y = \Big\| \sum_{i=1}^{n} (Hy)_i \phi(x_i) \Big\|^2.$$

But

$$Hy = y - \bar{y}\,1, \qquad \bar{y} := \frac{1}{n} \sum_{i=1}^{n} y_i.$$

Thus

$$\sum_i (Hy)_i \phi(x_i) = \sum_i (y_i - \bar{y}) \phi(x_i) = \sum_i y_i \phi(x_i) - \bar{y} \sum_i \phi(x_i) = \sum_i y_i (\phi(x_i) - \bar{\phi}) = \sum_i y_i \tilde{\phi}(x_i).$$

Therefore,

$$y^\top K_c y = \Big\| \sum_{i=1}^{n} y_i \tilde{\phi}(x_i) \Big\|^2 = y^\top \tilde{K} y,$$

where

$$\tilde{K}_{ij} = \langle \tilde{\phi}(x_i), \tilde{\phi}(x_j) \rangle = \langle \phi(x_i) - \bar{\phi}, \phi(x_j) - \bar{\phi} \rangle.$$

Since two symmetric matrices with equal quadratic forms for all y must be equal, we conclude

$$\boxed{K_c = \tilde{K}, \qquad K_c(i,j) = \langle \phi(x_i) - \bar{\phi},\, \phi(x_j) - \bar{\phi} \rangle.}$$

Thus **centering the kernel matrix is exactly equivalent to replacing the features by their mean-centered versions**.

Problem 5. (Eigenvalues of the Hat Matrix.) For kernel ridge regression define $H = K(K + \lambda I)^{-1}$.

(a) Show that H is symmetric.

(b) Recall that you can diagonalize symmetric PSD matrices like K as $K = Q \Lambda Q^\top$. Show its eigenvalues lie in $(0, 1)$.

Solution. Since K is symmetric, $H^\top = (K+\lambda I)^{-1}K = K(K+\lambda I)^{-1} = H$. Diagonalize $K = Q\Lambda Q^\top$ with $\Lambda = \operatorname{diag}(\lambda_i)$. Then

$$H = Q \operatorname{diag}\Big(\frac{\lambda_i}{\lambda_i + \lambda}\Big) Q^\top,$$

so eigenvalues are $\lambda_i/(\lambda_i + \lambda) \in (0,1)$.

Problem 6. (Gradient Check.) For $J(\alpha) = \|y - K\alpha\|^2 + \lambda\alpha^\top K\alpha$:

(a) Compute $\nabla_\alpha J(\alpha)$.

(b) Verify that setting it to zero gives $\alpha = (K + \lambda I)^{-1}y$.

Solution.

(a) Computing the gradient with respect to α, we have that

$$\nabla_\alpha J = -2K(y - K\alpha) + 2\lambda K\alpha = 2K(K + \lambda I)\alpha - 2Ky.$$

(b) Setting to zero yields $(K + \lambda I)\alpha = y$, the standard dual condition.

Problem 7. (Early Stopping as Regularization.) Gradient descent on $f(\theta) = \frac{1}{2}\|X\theta - y\|^2$ with step size η and $\theta_0 = 0$ yields

$$\theta_t = \eta \sum_{k=0}^{t-1} (I - \eta X^\top X)^k X^\top y.$$

(a) Show that

$$\sum_{k=0}^{t-1} R^k = (I - R^t)(I - R)^{-1} = (I - R)^{-1}(I - R^t)$$

(b) Use the fact in part (a) to show that

$$\theta_t = (I - (I - \eta X^\top X)^t)(X^\top X)^{-1}X^\top y.$$

(c) Diagonalize $X^\top X = Q\Lambda Q^\top$ and express

$$\theta_t = Q \operatorname{diag}\Big(\tfrac{1-(1-\eta\lambda_i)^t}{\lambda_i}\Big) Q^\top X^\top y.$$

(d) Explain how each factor $\frac{1-(1-\eta\lambda_i)^t}{\lambda_i}$ acts as a regularization filter.

Solution.

(a) Multiply both sides of the equation by $I - R$ to get the desired result:

$$\begin{aligned}\sum_{k=0}^{t-1} R^k (I - R) &= \sum_{k=0}^{t-1} R^k - \sum_{k=0}^{t-1} R^{k+1} \\ &= R^0 - R^t \\ &= I - R^t.\end{aligned}$$

And we have that

$$(I - R^t)(I - R)^{-1}(I - R) = I - R^t.$$

(b) Let $A = X^\top X$ and $R = I - \eta A$. Then

$$\theta_t \;=\; \eta \sum_{k=0}^{t-1} R^k X^\top y.$$

Therefore

$$\theta_t = \eta\,(I - (I-\eta A)^t)(\eta A)^{-1} X^\top y = (I - (I-\eta A)^t) A^{-1} X^\top y,$$

i.e.

$$\boxed{\theta_t \;=\; (I - (I - \eta X^\top X)^t)(X^\top X)^{-1} X^\top y.}$$

(c) Substitute $A = X^\top X = Q\Lambda Q^\top$ so that

$$\begin{aligned}
\theta_t &= (I - (I - \eta Q\Lambda Q^\top)^t)(Q\Lambda Q^\top)^{-1} X^\top y \\
&= (I - (QQ^\top - \eta Q\Lambda Q^\top)^t)(Q\Lambda Q^\top)^{-1} X^\top y \\
&= (I - Q(I - \eta\Lambda)^t Q^\top)(Q\Lambda Q^\top)^{-1} X^\top y
\end{aligned}$$

where the last step follows from the fact that Q is orthogonal. Indeed,

$$(Q\Lambda Q^\top)^t = Q\Lambda Q^\top Q\Lambda Q^\top \cdots Q\Lambda Q^\top = Q\Lambda^t Q^\top$$

Then we have that

$$\begin{aligned}
\theta_t &= (I - Q(I - \eta\Lambda)^t Q^\top)(Q\Lambda Q^\top)^{-1} X^\top y \\
&= \Big((Q\Lambda Q^\top)^{-1} - Q(I - \eta\Lambda)^t Q^\top (Q\Lambda Q^\top)^{-1}\Big) X^\top y \\
&= Q\,\mathrm{diag}\Big(\tfrac{1-(1-\eta\lambda_i)^t}{\lambda_i}\Big) Q^\top X^\top y.
\end{aligned}$$

Each eigen-direction is scaled by $f_t(\lambda_i) = \frac{1-(1-\eta\lambda_i)^t}{\lambda_i}$.

(d) For small λ_i, $f_t(\lambda_i)$ is small—these directions are suppressed. As $t \to \infty$, $f_t(\lambda_i) \to 1/\lambda_i$, recovering the unregularized least-squares solution.